Statement of Qualifications
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Describe your experience with enterprise-level infrastructure systems, including design, implementation, and maintenance. Highlight any work you've done with Active Directory, virtualization platforms, storage area networks (SANs), or cloud environments such as Azure, and AWS.

My knowledge and experience primarily come from creating Windows/Linux VM's using type 2 hypervisors such as Hyper-V, VMware, and VirtualBox. I experimented with the creation of Windows Server 2022 to get a feel for Active Directory, and how that ties in with DHCP/DNS services. I created a scenario where two Domain Controllers were configured for fail-over to simulate redundancy. I created and deleted accounts, set password requirements, reset passwords, and tested it all with Win 10/11 hosts and Ubuntu Desktop VMs. I felt that having basic knowledge of Active Directory was essential for me due to its widespread use amongst businesses, and to understand how core services remain available in larger environments when redundancy is enforced.

At one point I discovered Tailscale and this was a game changer for me. Any device that authenticates into your Tailnet can directly communicate using Tailscale overlay IPs, even if they're on different networks or locations. So now when I'm away from home, I can connect to my NAS where I host my media server, SMB for file-sharing, as well as an exit node that serves as a VPN. On the cloud side of things, I created AWS EC2 instances using my own template to deploy Tailscale exit nodes. These nodes served no practical use other than to help me develop some familiarity with the AWS platform as well as satisfying my own curiosity.

Explain your experience with IT security management and how you've contributed to protecting infrastructure environments. Include examples of how you've implemented security policies, worked with multi-factor authentication, or coordinated with security teams to meet compliance requirements.

While working on my home lab projects, I always tried to keep an emphasis on security. A great example I can give is with my Ubuntu Server that hosts all my services. I make sure to use key-based SSH authentication, specifically ed25519, disabling root login, and ensuring that only devices connected to my Tailnet can log in for remote management. Another thing I do is create a pfSense firewall that all my VMs can sit behind. Rules would be in place to create Layer 3 segmentation between my physical home network and the VMs behind the firewall. This was verified by pinging my main PC before and after setting the rule. Rules are also in place for allowing inbound SSH

sessions. In a live environment, these kinds of exercises will prove useful since enterprises follow similar procedures. These practices allow me to gain an understanding of foundational security concepts like controlled access and reducing attack surfaces. Concepts learned from these exercises share a very sharp parallel to real-world IT security management.

Provide an example of a complex IT project you managed or contributed to. Describe your role, the technologies involved, how you ensured successful delivery, and how the project supported broader organizational goals.

For my final Capstone project at WGU, I was required to simulate an enterprise level network for a fictional corporation using Cisco Packet Tracer. The corporation had a list of requirements with varying degrees of difficulty which I was required to satisfy in the network diagram and write-up. Concepts such as L2/L3 switching and routing, IPsec, DNS/DHCP, VLANs, and inter-VLAN routing needed to be demonstrated, and I was responsible for constructing the network diagram and functionality report. In this report, I referred to the corporation's requirements and how I planned to meet them with the network proposal.

The next step involved the implementation, testing, and validation of all the necessary requirements. I made sure to select proper routing paths whether they were static or dynamic, such as OSPF, between L2 and L3 devices. Testing and verification of the DNS/DHCP servers was conducted using manual DNS entries for local network navigation which were then tested on DHCP clients to verify connectivity. ACLs were implemented on the L3 switches to control inter-VLAN routing between hosts in different subnets. Rules were written to allow all subnets internet access, and limited access to internal resources based on roles/needs. Lastly, an IPsec site-to-site tunnel was created to allow encrypted round-trip communications to a simulated offsite location. Thorough documentation was necessary at each step to ensure the network’s behavior was clean and easily repeatable.

This project did well at demonstrating broader organizational goals in the sense that it allowed me to realize the importance of secure and efficient networking, and that good documentation supports scalability as an organization grows. Efficient routing, proper access controls, and encrypted communications serve not only to protect corporate assets but also to ensure usability for end users and devices.